What is Controller Manager?

Professional Answer

The Controller Manager runs control loops that continuously monitor cluster state and ensure the actual state matches the desired state.

Unprofessional Answer

Controller manages Kubernetes.

36. Examples of Kubernetes controllers?

Professional Answer

- Node Controller
- ReplicaSet Controller
- Deployment Controller
- Endpoint Controller

Unprofessional Answer

Deployment controller.

What does kube-controller-manager do internally?
kube-controller-manager runs multiple controllers that manage different cluster resources.
Each controller continuously compares the desired state with the actual state.
If a mismatch is detected, corrective actions are taken automatically.
Examples include ReplicaSet, Node, Deployment, and Endpoint controllers.
This reconciliation mechanism enables Kubernetes self-healing behavior.

4. kube-controller-manager (Reconciliation Engine)

Internal Design

- Runs **multiple controllers**
- Each controller is a **control loop**

Control Loop Logic

1. Watch desired state (API server)
2. Compare with actual state
3. Take corrective action

4. Repeat forever

Key Controllers Internals

- **ReplicaSet Controller**
  - Creates/deletes pods
- **Node Controller**
  - Detects node failure
- **Deployment Controller**
  - Manages rolling updates
- **Endpoint Controller**
  - Updates service endpoints

Core Principle

Kubernetes is a **reconciliation-based system**

Alright, now we're at the **heart of Kubernetes automation** 💓 — **kube-controller-manager**.
This is **classic senior DevOps interview territory**, so I'll go **deep, clean, and production-real**.

⚙️ **kube-controller-manager – Kubernetes Automation Engine**

1. What is kube-controller-manager?

kube-controller-manager is a **control-plane component** that runs multiple **controllers**.
Controllers continuously watch cluster state and **reconcile actual state to desired state**.
It operates on the **control loop pattern**.
It does not run workloads.
It ensures Kubernetes is self-healing.

2. Why kube-controller-manager exists

Kubernetes is **declarative**, not imperative.
Users define **desired state**.
Controllers enforce that state continuously.
Without controllers, Kubernetes would not auto-heal.
This component is the **brain of reconciliation**.

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3. How kube-controller-manager works (core loop)

Each controller follows this loop:

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1. Watch API Server
2. Compare desired vs current state
3. Take corrective action
4. Update API Server

This loop never stops.

4. Important controllers inside kube-controller-manager

1 Node Controller

- Monitors node health
- Marks nodes NotReady
- Evicts Pods when node is unhealthy

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2 Replication Controller

- Ensures correct Pod count
- Recreates deleted Pods
- Maintains availability

3 Deployment Controller

- Manages ReplicaSets
- Handles rolling updates
- Ensures deployment strategy

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4 StatefulSet Controller

- Maintains Pod identity
- Handles ordered deployment
- Manages stable storage

5 DaemonSet Controller

- Ensures one Pod per node
- Handles node join/leave
- Used for agents (logging, monitoring)

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6 Job & CronJob Controller

- Manages batch workloads
- Handles retries and failures
- Schedules periodic jobs

7 Endpoint Controller

- Updates Service endpoints
- Tracks healthy Pods
- Keeps traffic routing accurate

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8 Namespace Controller

- Manages namespace lifecycle
- Handles deletion & cleanup
- Removes dependent resources

9 ServiceAccount & Token Controller

- Creates default service accounts
- Manages token secrets
- Enables Pod authentication

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5. kube-controller-manager vs Scheduler

kube-controller-manager	kube-scheduler
Maintains desired state	Picks node
Continuous reconciliation	One-time decision
Fixes drift	Initial placement
Multiple controllers	Single function

6. What happens if kube-controller-manager goes down?

Feature	Impact
Existing Pods	Continue running
Auto-healing	Stops
Scaling	Stops

Feature	Impact
Node eviction	Stops
Jobs	Fail silently

👉 Cluster becomes **non self-healing**.

7. Real production outage example

Incident

- Controller Manager crashed due to config error

Symptoms

- Pods not recreated after deletion
- Nodes marked NotReady but Pods still running
- Deployments stuck

Root cause

Controller loops stopped

Fix

- Restarted kube-controller-manager
- Validated flags & certificates

8. Leader election in kube-controller-manager

- Multiple instances can run
- Only one active leader
- Uses **etcd-based leader election**
- Prevents duplicate actions

👉 HA without double execution.

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9. How kube-controller-manager is deployed

- Runs as **static pod**
- Or systemd service
- Communicates with API Server
- Does not talk to etcd directly

10. Security model

- Authenticates to API Server
- Uses RBAC permissions
- Restricted access
- No direct node access

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11. Common kube-controller-manager issues

- Certificate expiry
- Leader election failures
- High API latency
- Misconfigured flags
- API Server unavailable

12. How to troubleshoot kube-controller-manager

kubectl get pods -n kube-system

kubectl logs kube-controller-manager-<node>

kubectl describe pod kube-controller-manager-<node>

Check:

- Leader election logs
- Controller failures
- API connectivity

13. Ultra-hard interview question

Q:
Pods are deleted but not recreated. Scheduler is healthy. Why?

Answer:
kube-controller-manager is likely down.
Scheduler only places Pods.
Controllers create Pods.
Without controller manager, desired state is not enforced.

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14. Professional vs Unprofessional answer

Unprofessional:

Controller manager controls stuff.

Professional:

kube-controller-manager runs multiple reconciliation controllers that continuously enforce the desired cluster state by monitoring Kubernetes objects and triggering corrective actions via the API Server.

15. Exam one-liner

kube-controller-manager ensures Kubernetes is self-healing by reconciling desired and actual states.

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16. Senior-level red flags

- 🚩 "Controller Manager schedules Pods"
- 🚩 "Controller Manager stores data"
- 🚩 "Controller Manager talks directly to nodes"

(All false.)

17. Golden rule

Scheduler places Pods, controller-manager keeps them alive.